MD ASHIKUL HAQUE

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Google Scholar: scholar.google.com/citations?user=11vTHPYAAAAJ

EDUCATION

PhD, Computer Science Expected May 2027
University of Texas at Dallas, USA
Dissertation: *Dependable Low-Power Wide-Area Networking under Contention: Jamming, Coexistence, and Deadlines*
Advisor: **Prof. Abusayeed Saifullah**

MSc, Computer Science 2024
Wayne State University, USA

BSc, Computer Science and Engineering 2018
Bangladesh University of Engineering and Technology, Bangladesh

RESEARCH INTERESTS

Anti-Jamming and Physical-Layer Security for Low-Power Wide-Area Networks (LPWANs) • Coexistence among Learning-Enabled Networks • Real-Time and Mixed-Criticality Scheduling • Cross-Technology and Direct-to-Satellite Links • Deadline-Aware Scheduling for Large Language Model (LLM) Agents and Connected Vehicles

Research Summary. I build low-power wireless systems that stay correct and on time when the resource they depend on is contested by an attacker, by neighboring networks, or by a deadline. Defense against jamming and coexistence with other networks are very challenging in LPWANs for the Internet of Things (IoT). Solutions designed for other wireless networks cannot be adopted directly, because LPWAN devices have limited computation, low transmission power, and small batteries. My work covers physical-layer defenses that keep LoRa networks decoding under jamming attacks, learning-based coexistence management for dense LPWAN deployments, medium access control (MAC) and mixed-criticality scheduling for industrial wireless networks, cross-technology communication from Long Range Frequency Hopping Spread Spectrum (LR-FHSS) to LoRa, and deadline-aware token scheduling for LLM agents that drive physical actuators. I validate these designs on software-defined radio (SDR) prototypes with GNU Radio and USRP B200, on commercial off-the-shelf (COTS) LoRa hardware and multi-node testbeds, and in PyTorch, NS-3, and CARLA at scale.

CONFERENCE PUBLICATIONS

Each paper's title links to its full text.

[C9] **Md Ashikul Haque**, Abusayeed Saifullah. *Deadline-Aware Token Scheduling for Physical-I/O LLM Agents*. In Proc. IEEE Real-Time Systems Symposium (RTSS 2026). Acceptance rate: 14% (50 of 350).

[C8] **Md Ashikul Haque**, Aakriti Jain, Venkata Prashant Modekurthy, Abusayeed Saifullah. *Enabling Cross Technology Communication from LR-FHSS to LoRa*. In Proc. ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys 2026), CPS-IoT Week 2026. Acceptance rate: 19.8% (52 of 263).

[C7] **Md Ashikul Haque**, Abusayeed Saifullah. *Decoding LoRa Packets under Collaborative Jamming Attacks*. In Proc. 23rd International Conference on Embedded Wireless Systems and Networks (EWSN 2026). Acceptance rate: 21% (15 of 71).

[C6] **Md Ashikul Haque**, Abusayeed Saifullah. *ReMix: Resource-Reclaiming Mixed-Criticality Scheduling for Industrial IoT*. In Proc. The 35th International Conference on Computer Communications and Networks (ICCCN 2026). Acceptance rate: 25%.

[C5] **Md Ashikul Haque**, Abusayeed Saifullah. *Mitigating Jamming Attacks in LoRa Networks: A Defense Strategy against LoRa-Based Jammers*. In Proc. ACM International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc 2025). Acceptance rate: 23% (39 of 169).

[C4] **Md Ashikul Haque**, Abusayeed Saifullah, Haibo Zhang. *Deep Reinforcement Learning Based Coexistence Management in LPWAN*. In Proc. IEEE Conference on Computer Communications (INFOCOM 2025). Acceptance rate: 18.7% (272 of 1458).

[C3] **Md Ashikul Haque**, Abusayeed Saifullah. *Handling Jamming Attacks in a LoRa Network*. In Proc. ACM/IEEE Conference on Internet of Things Design and Implementation (IoTDI 2024), CPS-IoT Week 2024.

[C2] Aakriti Jain, **Md Ashikul Haque**, Abusayeed Saifullah, Haibo Zhang. *Burst-MAC: A MAC Protocol for Handling Burst Traffic in LoRa Network*. In Proc. IEEE Real-Time Systems Symposium (RTSS 2024).

[C1] **Md Ashikul Haque**, Abusayeed Saifullah. *A Game-Theoretic Approach for Mitigating Jamming Attacks in LPWAN*. In Proc. International Conference on Embedded Wireless Systems and Networks (EWSN 2023). To our knowledge, the first anti-jamming work in LPWAN.

Workshop Publications

[W2] **Md Ashikul Haque**, Abusayeed Saifullah, Haibo Zhang. *Coordinating Coexisting Learning Agents in Shared Spectrum via Parameter Space Complementarity*. Agents in the Wild: Safety, Security, and Beyond (ICLR 2026).

[W1] **Md Ashikul Haque**, Dali Ismail, Abusayeed Saifullah. *Scalable Real-Time Control in Industrial Cyber-Physical Systems*. In Proc. Workshop on Software-Defined Networking for Cyber-Physical System Automation (SDN-CPS 2024), co-located with ICDCN 2024, pp. 353–358.

Selected Talks

ReMix: Resource-Reclaiming Mixed-Criticality Scheduling for Industrial IoT — Jul 2026
IEEE ICCCN 2026, Honolulu, Hawaii, USA

Mitigating Jamming Attacks in LoRa Networks: A Defense Strategy against LoRa-Based Jammers — Oct 2025
ACM MobiHoc 2025, Rice University, Houston, USA

Handling Jamming Attacks in a LoRa Network — May 2024
ACM/IEEE IoTDI 2024, Hong Kong, China

Research and Professional Experience

Graduate Research Assistant, Dept. of Computer Science — 2022 to 2026
Wayne State University, USA (Fall 2022 to Summer 2025) and The University of Texas at Dallas, USA (Fall 2025 and Summer 2026) | Advisor: **Prof. Abusayeed Saifullah**

- Published 9 conference papers and 2 workshop papers in wireless networking and real-time systems, 10 of them as first author.
- Designed gateway-side and network-server-side defenses that recover LoRa packets under single, LoRa-based, collaborative, and spatially distributed jammers, with outdoor evaluation on USRP B200 gateways and commodity LoRa nodes.
- Built learning-based coexistence management at the LoRa Network Server and disclosure-free coordination among LoRaWAN operators, evaluated in PyTorch, NS-3, and a demodulator-level simulation of 32,000 devices across 8 operators.
- Designed real-time medium access and mixed-criticality scheduling for LoRa and industrial wireless sensor-actuator networks, evaluated on LoRa devices and a 15-node Tmote Sky testbed, and extended the same scheduling method to LLM serving on vLLM/ROCm and to cooperative perception on a three-car MentorPi testbed and CARLA.
- Built cross-technology communication from LR-FHSS to LoRa, per-message delivery certificates, and signature-bound frame recovery for direct-to-satellite LR-FHSS on GNU Radio and USRP.

Assistant Engineer, ICT Division — 2020 to 2021
Dhaka Electric Supply Company, Bangladesh

- Maintained ICT infrastructure with monitoring, incident response, and reliability focused operations for distributed utility systems.

Software Engineer, IoT Division — 2019 to 2020
Samsung R&D Institute Bangladesh, Bangladesh

- Worked across both the user interface and the cloud integration layers of the SmartThings iOS application, which Samsung reported serving more than 45 million monthly active users and 5,000 supported device types in October 2019.
- Developed production-grade C++ components in the Samsung Cloud Client (SCClient), the shared library that handles cloud connectivity and device communication for SmartThings, and exposed them to iOS through a Swift wrapper.
- Built the device onboarding dashboard through which users register, authenticate, and configure new devices.
- Automated the entire build, packaging, and release workflow of the SCClient iOS team in Bash, which removed the manual steps from every release and made each build reproducible.

Teaching

Graduate Teaching Assistant, Dept. of Computer Science Spring and Fall 2026
The University of Texas at Dallas, USA
Courses: CS 5390 Computer Networks; CS 6352 Performance of Computer Systems and Networks

- Graded assignments, exams, and project deliverables for both courses.
- Helped students understand course concepts through office hours, question answering, and discussion of networking and systems topics.
- Supported course projects by clarifying requirements, troubleshooting implementations, and providing feedback on design and debugging.

Guest Lecturer, Dept. of Computer Science 2025
The University of Texas at Dallas, USA
Course: Internet of Things (graduate). Delivered multiple lectures on LoRa networking and network simulation in NS-3.

Lecturer, Dept. of Computer Science 2018 to 2019
Millennium University, Bangladesh
Courses: Structured Programming; Data Communication. Instructor of record for both courses: designed lectures, laboratory assignments, and examinations.

Mentoring

Nasif Ahmed, PhD student, The University of Texas at Dallas January 2025 to present
Junior doctoral student in our laboratory. I mentor him on identifying new problems and approaching their solutions, on experiments with USRP radios and commercial off-the-shelf (COTS) LoRa devices, and on writing systems papers.

Awards and Honors

Dean's List Award, Wayne State University 2024
Professional Travel Award, College of Engineering, Wayne State University 2024
ACM SIGBED Travel Grant, CPS-IoT Week, San Antonio, Texas, USA 2023

Research Funding and Proposals

- Contributed to the writing of an NSF proposal on real-time cooperative perception for connected vehicles, with **Prof. Abusayeed Saifullah** as principal investigator 2026
- My doctoral research has been supported by five NSF awards (CNS-2301757, CAREER-2306486, CNS-2306745, CNS-2601685, and CNS-2602744) and one ONR award (N00014-23-1-2151), all with **Prof. Abusayeed Saifullah** as principal investigator 2022 to present

Professional Service

Journal Reviewer

- ACM Transactions on Cyber-Physical Systems 2026
- Pervasive and Mobile Computing (Elsevier) 2023

Conference Sub-reviewer

- IEEE Intl. Conf. on Distributed Computing in Smart Systems and the IoT (DCOSS-IoT) 2026
- IEEE Intl. Conf. on Embedded and Real-Time Computing Systems and Applications (RTCSA) 2024, 2025
- IEEE Intl. Conf. on Sensing, Communication, and Networking (SECON) 2026

Conference Referee

- ACM/IEEE Intl. Conf. on Embedded Artificial Intelligence and Sensing Systems (SenSys) 2022, 2024, 2025, 2026
- IEEE Real-Time Systems Symposium (RTSS) 2023, 2024, 2025, 2026
- IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS) 2023, 2024
- Intl. Conf. on Embedded Wireless Systems and Networks (EWSN) 2023, 2024, 2025
- IEEE Intl. Conf. on Distributed Computing in Smart Systems and the IoT (DCOSS-IoT) 2023, 2024, 2025

Journal Referee

- IEEE/ACM Transactions on Networking 2024, 2025
- ACM Transactions on Cyber-Physical Systems 2025

Other Service

- Student Volunteer, CPS-IoT Week 2023, San Antonio, Texas, USA 2023

Technical Skills

Programming:	C, C++, Python, MATLAB, SQL, Swift, Bash
Wireless and SDR:	GNU Radio, USRP B200, LoRa and LR-FHSS physical layers, LoRaWAN, COTS LoRa nodes and gateways, Tmote Sky
Learning and simulation:	PyTorch, Pandas, NS-3, CARLA, vLLM
Systems and tools:	Linux, Docker, Git, LaTeX

References

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